

WORK №5. Optimal observer (Kalman filter)

Given the plant

$$\begin{aligned}\dot{x} &= Ax + bu + Gw, \quad x(0), \\ y &= Cx + v,\end{aligned}$$

where w, v — “white noises” with zero expected values $M\{w\} = M\{v\} = 0$ and autoregressive functions $M\{w(t)w^T(\tau)\} = W\delta(t - \tau)$, $M\{v(t)v(\tau)\} = V\delta(t - \tau)$ with known constant spectral density W and V respectively.

The objective is to design an optimal observer generating estimates $\hat{x}$:

$$\|x(t) - \hat{x}(t)\| \leq \Delta \quad \forall t \geq T,$$

where Δ and T are the maximum steady-state error and the transient time, respectively. Cost function is defined as follows:

$$J = M\{e_H^T e_H\}$$

where $e_H = x - \hat{x}$ is observer error, $M\{\cdot\}$ is expected value.

The observer is defined by the following structure:

$$\dot{\hat{x}} = A\hat{x} + bu + L(y - C\hat{x}) \quad \hat{x}(0),$$

where matrix L is calculated with the Riccati equation

$$\begin{aligned}AP + PA^T + GWG^T - PC^T V^{-1} CP &= 0, \\ L &= PC^T V^{-1},\end{aligned}$$

The order of the work

1. Based on the known matrices A, b , shown in the table, matrix $C = [1 \ 0]$, matrix $G = I$, and also W и V calculate the matrix L .
2. Simulate a closed-loop system with initial conditions $x(0) = [1, 0]^T$ and $u = \sin t$. Plot the variables e_{H1}, e_{H2} .
3. Negligibly change L parameters so that the system preserves the stability and repeat № 2 with the same simulation time. Compare with results obtained in № 2 and make a conclusion.
4. Change W parameters so that the matrix W remains symmetric and positive-defined and repeat № 2 with the same simulation time. Compare with results obtained in № 2 and make a conclusion.

5. Change V so that the value V remains positive and repeat № 2 with the same simulation time. Compare with results obtained in № 2 and make a conclusion.

№	A, b	W	V	№	A, b	W	V
1	$\begin{bmatrix} 0 & 1 \\ -8 & -4 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix}$	$\begin{bmatrix} 4 & 5 \\ 5 & 7 \end{bmatrix}$	1	26	$\begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 3 & 2 \\ 2 & 4 \end{bmatrix}$	1
2	$\begin{bmatrix} 0 & 1 \\ -5 & -7 \end{bmatrix} \begin{bmatrix} 0 \\ 4 \end{bmatrix}$	$\begin{bmatrix} 6 & 3.5 \\ 3.5 & 4 \end{bmatrix}$	2	27	$\begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 2 & 1.5 \\ 1.5 & 2 \end{bmatrix}$	2
3	$\begin{bmatrix} -4 & 1 \\ 0 & -3 \end{bmatrix} \begin{bmatrix} 6 \\ 4 \end{bmatrix}$	$\begin{bmatrix} 3 & 4 \\ 4 & 9 \end{bmatrix}$	3	28	$\begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix}$	$\begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix}$	3
4	$\begin{bmatrix} -5 & 0 \\ 0 & -5 \end{bmatrix} \begin{bmatrix} 6 \\ 2 \end{bmatrix}$	$\begin{bmatrix} 3 & 2 \\ 2 & 2 \end{bmatrix}$	4	29	$\begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 3 & 2 \\ 2 & 2 \end{bmatrix}$	4
5	$\begin{bmatrix} 0 & -4 \\ 1 & -4 \end{bmatrix} \begin{bmatrix} 8 \\ 7 \end{bmatrix}$	$\begin{bmatrix} 4 & 3.5 \\ 3.5 & 5 \end{bmatrix}$	5	30	$\begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \end{bmatrix}$	$\begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}$	5
6	$\begin{bmatrix} 0 & -9 \\ 1 & -4 \end{bmatrix} \begin{bmatrix} 5 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 7 & 5 \\ 5 & 6 \end{bmatrix}$	1	31	$\begin{bmatrix} 0 & -6 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix}$	1
7	$\begin{bmatrix} 0 & 1 \\ -4 & -3 \end{bmatrix} \begin{bmatrix} 7 \\ 3 \end{bmatrix}$	$\begin{bmatrix} 5 & 4.5 \\ 4.5 & 6 \end{bmatrix}$	2	32	$\begin{bmatrix} 0 & 1 \\ -1 & -5 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 2 & 2.5 \\ 2.5 & 4 \end{bmatrix}$	2
8	$\begin{bmatrix} -5 & 1 \\ 0 & -4 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix}$	$\begin{bmatrix} 3 & 4 \\ 4 & 9 \end{bmatrix}$	3	33	$\begin{bmatrix} -2 & 1 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 1 \\ 1 & 7 \end{bmatrix}$	3
9	$\begin{bmatrix} 0 & 1 \\ -4 & -6 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix}$	$\begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix}$	4	34	$\begin{bmatrix} 0 & 1 \\ -1 & -5 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix}$	$\begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$	4
10	$\begin{bmatrix} 0 & -7 \\ 1 & -6 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix}$	$\begin{bmatrix} 8 & 5 \\ 5 & 6 \end{bmatrix}$	5	35	$\begin{bmatrix} 0 & -2 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 6 & 3 \\ 3 & 2 \end{bmatrix}$	5
11	$\begin{bmatrix} -1 & 0 \\ 0 & -6 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix}$	$\begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$	7	36	$\begin{bmatrix} -5 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \end{bmatrix}$	$\begin{bmatrix} 10 & 2 \\ 2 & 1 \end{bmatrix}$	7
12	$\begin{bmatrix} 0 & -2 \\ 1 & -7 \end{bmatrix} \begin{bmatrix} 8 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 9 & 5 \\ 5 & 3 \end{bmatrix}$	1	37	$\begin{bmatrix} 0 & -3 \\ 1 & -6 \end{bmatrix} \begin{bmatrix} 5 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 7 & 2 \\ 2 & 6 \end{bmatrix}$	1
13	$\begin{bmatrix} 0 & 1 \\ -7 & -6 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 9 & 7.5 \\ 7.5 & 8 \end{bmatrix}$	2	38	$\begin{bmatrix} 0 & 1 \\ -4 & -7 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix}$	$\begin{bmatrix} 7 & 4.5 \\ 4.5 & 6 \end{bmatrix}$	2
14	$\begin{bmatrix} -5 & 1 \\ 0 & -9 \end{bmatrix} \begin{bmatrix} 6 \\ 5 \end{bmatrix}$	$\begin{bmatrix} 9 & 4 \\ 4 & 6 \end{bmatrix}$	3	39	$\begin{bmatrix} -3 & 1 \\ 0 & -4 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 3 & 4 \\ 4 & 8 \end{bmatrix}$	3

15	$\begin{bmatrix} 0 & 1 \\ -2 & -8 \end{bmatrix} \begin{bmatrix} 9 \\ 7 \end{bmatrix}$	$\begin{bmatrix} 3 & 1 \\ 1 & 7 \end{bmatrix}$	4	40	$\begin{bmatrix} 0 & 1 \\ -3 & -2 \end{bmatrix} \begin{bmatrix} 3 \\ 8 \end{bmatrix}$	$\begin{bmatrix} 9 & 4 \\ 4 & 3 \end{bmatrix}$	4
16	$\begin{bmatrix} -9 & 1 \\ 0 & -6 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \end{bmatrix}$	$\begin{bmatrix} 7 & 4 \\ 4 & 5 \end{bmatrix}$	3	41	$\begin{bmatrix} 0 & 1 \\ -3 & -4 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix}$	$\begin{bmatrix} 3 & 4 \\ 4 & 7 \end{bmatrix}$	4
17	$\begin{bmatrix} 0 & 1 \\ -6 & -7 \end{bmatrix} \begin{bmatrix} 5 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 9 & 8 \\ 8 & 9 \end{bmatrix}$	4	42	$\begin{bmatrix} 0 & -5 \\ 1 & -5 \end{bmatrix} \begin{bmatrix} 3 \\ 3 \end{bmatrix}$	$\begin{bmatrix} 5 & 2 \\ 2 & 3 \end{bmatrix}$	5
18	$\begin{bmatrix} 0 & -9 \\ 1 & -9 \end{bmatrix} \begin{bmatrix} 3 \\ 7 \end{bmatrix}$	$\begin{bmatrix} 7 & 5 \\ 5 & 4 \end{bmatrix}$	5	43	$\begin{bmatrix} -3 & 0 \\ 0 & -5 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix}$	$\begin{bmatrix} 6 & 6 \\ 6 & 8 \end{bmatrix}$	7
19	$\begin{bmatrix} -3 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 9 \end{bmatrix}$	$\begin{bmatrix} 9 & 6 \\ 6 & 5 \end{bmatrix}$	7	44	$\begin{bmatrix} 0 & -8 \\ 1 & -4 \end{bmatrix} \begin{bmatrix} 4 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 7 & 2 \\ 2 & 6 \end{bmatrix}$	1
20	$\begin{bmatrix} 0 & -4 \\ 1 & -6 \end{bmatrix} \begin{bmatrix} 4 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 8 & 3 \\ 3 & 2 \end{bmatrix}$	1	45	$\begin{bmatrix} 0 & 1 \\ -3 & -6 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix}$	$\begin{bmatrix} 7 & 5 \\ 5 & 9 \end{bmatrix}$	2
21	$\begin{bmatrix} 0 & 1 \\ -3 & -4 \end{bmatrix} \begin{bmatrix} 5 \\ 4 \end{bmatrix}$	$\begin{bmatrix} 9 & 7 \\ 7 & 6 \end{bmatrix}$	2	46	$\begin{bmatrix} -5 & 1 \\ 0 & -8 \end{bmatrix} \begin{bmatrix} 6 \\ 4 \end{bmatrix}$	$\begin{bmatrix} 8 & 3 \\ 3 & 2 \end{bmatrix}$	3
22	$\begin{bmatrix} -3 & 1 \\ 0 & -4 \end{bmatrix} \begin{bmatrix} 0 \\ 5 \end{bmatrix}$	$\begin{bmatrix} 3 & 4 \\ 4 & 6 \end{bmatrix}$	3	47	$\begin{bmatrix} 0 & 1 \\ -8 & -6 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix}$	$\begin{bmatrix} 11 & 3 \\ 3 & 2 \end{bmatrix}$	4
23	$\begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 9 \end{bmatrix}$	$\begin{bmatrix} 3 & 2 \\ 2 & 8 \end{bmatrix}$	4	48	$\begin{bmatrix} 0 & 1 \\ -9 & -2 \end{bmatrix} \begin{bmatrix} 8 \\ 4 \end{bmatrix}$	$\begin{bmatrix} 7 & 4 \\ 4 & 5 \end{bmatrix}$	4
24	$\begin{bmatrix} 0 & 1 \\ -6 & -6 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$	5	49	$\begin{bmatrix} 0 & 1 \\ -1 & -3 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix}$	$\begin{bmatrix} 3 & 1 \\ 1 & 4 \end{bmatrix}$	5
25	$\begin{bmatrix} -2 & 1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 7 \end{bmatrix}$	$\begin{bmatrix} 5 & 2 \\ 2 & 3 \end{bmatrix}$	6	50	$\begin{bmatrix} 0 & 1 \\ -5 & -8 \end{bmatrix} \begin{bmatrix} 1 \\ 6 \end{bmatrix}$	$\begin{bmatrix} 7 & 2 \\ 2 & 8 \end{bmatrix}$	6